Many small bond changes close the HOMO–LUMO gap of phenylenes long before one large change does

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Abstract

Closing the Hückel HOMO–LUMO gap of biphenylene through a single bond requires that bond to shorten by about 1.8 Å, more than its own length; letting all fourteen bonds change, some up and some down, requires at most 0.12 Å per bond. We explain why many small changes win, how large they must be and where they must sit in phenylenes, whose Kekulé structures do not all share one sign. The gap closes when Kekulé structures of opposite sign cancel. The smallest relative change of the couplings that achieves this, the closure threshold ρ, is the regularity radius of an interval matrix and depends only on the signed Kekulé structures. When one structure carries the minority sign, ρ follows from $\sum_a q^{k_a} = 1$ with $q = (1-\rho)/(1+\rho)$, giving $q^2 = 1/\varphi$ for benzocyclobutadiene and $q^3 + q = 1$ for biphenylene; for [N]phenylenes with $N \leq 4$ we compute ρ exactly. Closure concentrates at one end of angular and branched phenylenes and spreads along linear ones. In a bond-length model, the closure energy is set by the squared gap over P, which measures how strongly the HOMO and LUMO partial bond orders differ on few bonds; g_0^2/P and ρ rank the closure energies of nine skeletons with Spearman correlations 0.97 and 0.94, against 0.72 for the relaxed gap. In a preregistered B3LYP test on biphenylene and angular [3]phenylene, the predicted pattern lowers the gap 2.4–3.6 times more efficiently than undoing bond alternation and retains about 80% of its model advantage over random patterns.

Keywords: Hückel theory; Kekulé structures; HOMO–LUMO gap; interval-matrix regularity; phenylenes; diradical character
Mathematics Subject Classification (2020): 92E10; 05C92; 15A18; 15A15

1 Introduction

To close the Hückel HOMO–LUMO gap of biphenylene by changing a single bond, one of the two bonds joining the benzene rings must have its coupling multiplied by four (Fig. 1a). With the usual linear dependence of coupling on bond length, that means shortening the bond by 1.8 Å, more than its own length. If every bond may change, the gap closes once each coupling has moved by 18.9%, some up and some down (Fig. 1b): at most 0.12 Å per bond. In energy the contrast is as stark. In the bond-length model of Section 5, closing the gap from the relaxed geometry through the single bond costs 113 eV; letting all bonds adjust costs 4.2 eV. This paper asks why many small changes win, how large they must be for a given carbon skeleton, and where in the molecule they have to sit.

The question matters because a shrinking gap pushes a molecule toward open-shell, diradical character, a central target in the chemistry of polycyclic hydrocarbons that contain four-membered rings [1]. A closed Hückel gap is the limit of this trend rather than its onset: in the density functional

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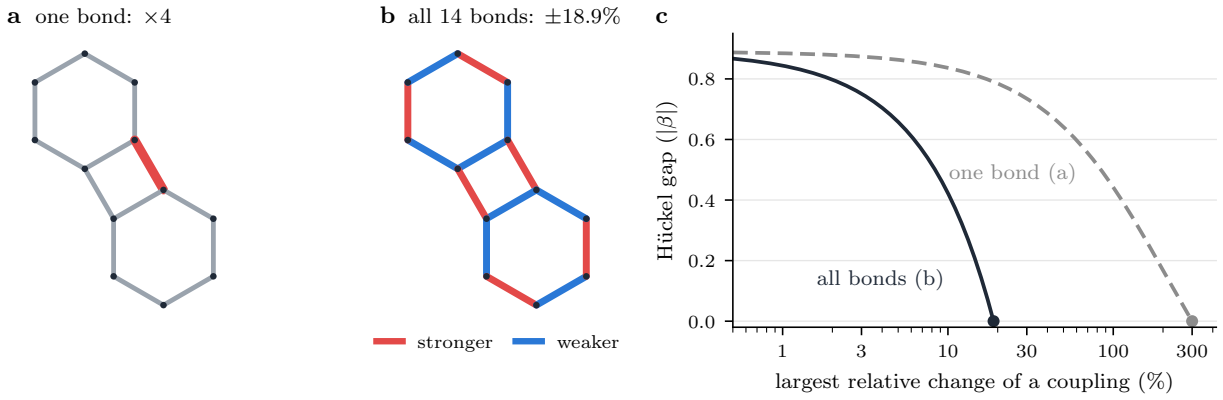


Figure 1: Closing the Hückel gap of biphenylene. (a) The smallest single-bond change: the coupling of the red bond must grow from 1 to 4. (b) The cheapest cooperative change: red couplings grow and blue couplings shrink, each by 18.9%. (c) The gap along both paths; the horizontal axis is the largest relative change of any coupling. With $t = t_0 - \alpha\,\delta$ ($t_0 = 2.5$ eV, $\alpha = 4.1$ eV Å^{-1}), path (a) shortens one bond by 1.8 Å and path (b) changes each bond by 0.12 Å.

calculations of Section 6 the singlet breaks spin symmetry while the gap is still about 2 eV. The phenylenes, which alternate benzene and cyclobutadiene rings, are the classic family: linear members become more reactive and absorb further to the red as they grow, while angular members stay more stable [2, 3]. Substituents, strain and fusion all act on the molecule by changing bond lengths, and through them the bond couplings. Which local changes push the gap to zero, and how far, is therefore a design question.

The size of the gap does not answer it. The gap says how far the frontier levels are from each other, not which bonds move them. Perturbation theory gives the first derivatives (the partial bond orders of Coulson [4]) and the second derivatives (the bond–bond polarizabilities, which underlie the theory of bond fixation and of symmetry lowering by π electrons [5–7]). These describe infinitesimal changes. Here we ask for the finite change that makes the gap vanish, and for where it must sit.

We measure the change in three ways, each answering one question. The largest relative change of a coupling, the closure threshold ρ, asks how far the gap is from closing in the space of couplings; it is fixed by the carbon skeleton and can be computed exactly. The energy in a bond-length model asks what closing costs. The gap decrease per unit energy in density functional theory asks whether the pattern predicted by the model works in real molecules. The three agree where they can be compared: ρ ranks the closure energies of the model with Spearman correlation 0.94, and the predicted pattern keeps most of its advantage under B3LYP.

2 Signed Kekulé structures and the closure threshold

We use the Hückel model of an alternant hydrocarbon with $2n$ carbon atoms, n in each of the two sublattices (starred and unstarred). With Coulomb integrals set to zero and couplings $t_{ij} = -\beta_{ij} > 0$ between bonded atoms i (starred) and j (unstarred), the Hamiltonian is

$$H = -\begin{pmatrix} 0 & T \\ T^{\mathsf{T}} & 0 \end{pmatrix}, \tag{1}$$

where T is the $n \times n$ matrix of couplings. The orbital energies are $\pm\sigma_k(T)$, the singular values of T, so the gap is

$$g = 2\,\sigma_{\min}(T), \tag{2}$$

and it closes exactly when $\det T = 0$. All couplings equal one unless stated; energies are in units of $|\beta|$.

The determinant counts Kekulé structures with signs. Expanding $\det T$ gives one term for each Kekulé structure, that is, each set of bonds that covers every atom exactly once:

$$\det T = \sum_M s_M \prod_{(ij)\in M} t_{ij}, \qquad s_M = \pm 1. \tag{3}$$

Two Kekulé structures differ by a set of disjoint cycles whose bonds alternate between them, and their signs differ exactly when an odd number of these cycles have length $4k$ [8]. With equal couplings, $|\det T|$ is the algebraic structure count: the number of Kekulé structures of one sign minus the number of the other [9, 10]; recurrence formulas compute it efficiently for any bipartite graph [11]. In benzenoid hydrocarbons every pair differs by $(4k+2)$-cycles, all terms share one sign, and $\det T$ stays positive for any positive couplings. The gap of benzene or naphthalene cannot be closed without removing a bond.

Four-membered rings bring terms of the opposite sign. Benzocyclobutadiene has three Kekulé structures and an algebraic structure count of one; biphenylene has five and three. For phenylenes the algebraic structure count equals the number of Kekulé structures of the hexagonal squeeze, the benzenoid obtained by squeezing out the four-membered rings [12], as the values in Table 1 illustrate (biphenylene and naphthalene: three; linear [3]phenylene and anthracene: four). The Kekulé structures of phenylenes have been studied in detail [13], and many of their π-electron properties correlate with those of the hexagonal squeeze [14]. The closure threshold does not: naphthalene, the squeeze of biphenylene, cannot be closed at all (Table 1). In benzocyclobutadiene and biphenylene, exactly one structure carries the minority sign: in biphenylene it is the structure in which both bonds joining the benzene rings are double, so that each benzene ring keeps only two double bonds.

Changing one bond is inefficient. Changing the coupling of bond (ij) by t changes T by a rank-one matrix, and

$$\det\Big(T + t\,e_i e_j^{\mathsf{T}}\Big) = \det T\,\big[1 + t\,(T^{-1})_{ji}\big]. \tag{4}$$

The gap closes at $t^* = -1/(T^{-1})_{ji}$. For biphenylene the smallest feasible value is $t^* = 3$, on either bond joining the benzene rings. One bond enters the terms of only some Kekulé structures and must carry the whole cancellation alone.

Changing every bond by a little is efficient. Every Kekulé structure contains n bonds, so a small relative change applied to all of them compounds in each term of Eq. (3). To make the minority term win, strengthen its bonds and weaken all others. We measure the size of a change by its largest relative component and define the *closure threshold*

$$\rho = \min\big\{\epsilon \ge 0 :\ \det(T+E) = 0 \text{ for some } E \text{ with } |E_{ij}| \le \epsilon\, T_{ij}\big\}. \tag{5}$$

Bonds that are absent stay absent, and for $\epsilon < 1$ every coupling stays positive. Because ρ is defined through $\det T$ alone, it depends only on the signed Kekulé structures: two skeletons whose Kekulé structures correspond bond by bond, with the same signs, have the same threshold even when their spectra differ. For a skeleton with one minority structure, ρ has a closed form.

Proposition 1. *Let the skeleton have K Kekulé structures and $\det T \neq 0$ at equal couplings. Let M_- be the only Kekulé structure whose sign differs from all others, let $M_1, \ldots, M_{K-1}$ be the others, and let $k_a = |M_a \setminus M_-|$, the number of bonds in which M_a differs from M_-. Then $\rho = (1-q)/(1+q)$, where q is the unique root in $(0,1)$ of*

$$\sum_{a=1}^{K-1} q^{k_a} = 1. \tag{6}$$

The root is unique because the left side increases strictly from 0 to $K-1 \geq 2$ on $[0,1]$.

Proof. Divide Eq. (3) by the minority term. Bonds shared by M_a and M_- cancel, so $|\det T| / \prod_{M_-} t = |\sum_a r_a - 1|$ with

$$r_a = \frac{\prod_{(ij)\in M_a \setminus M_-} t_{ij}}{\prod_{(ij)\in M_- \setminus M_a} t_{ij}}. \tag{7}$$

Both products contain k_a couplings. With every coupling in $[1-\epsilon, 1+\epsilon]$, $r_a \geq q^{k_a}$ with $q = (1-\epsilon)/(1+\epsilon)$, and the bound is reached by setting the couplings of M_- to $1+\epsilon$ and all others to $1-\epsilon$. This one setting reaches the bound for every a at once, so the smallest value of $\sum_a r_a$ over all allowed couplings is $\sum_a q^{k_a}$. The determinant can vanish only if this value is at most one, and it vanishes at the first ϵ where it equals one. □

For benzocyclobutadiene the two majority structures differ from the minority one in two bonds (around the four-membered ring) and in four bonds (around the eight-membered perimeter), so $q^2 + q^4 = 1$, $q^2 = 1/\varphi$ with φ the golden ratio, and

$$\rho_{\mathrm{BCB}} = \frac{\sqrt{\varphi} - 1}{\sqrt{\varphi} + 1} = 0.11973. \tag{8}$$

For biphenylene the exponents are 2, 4, 4 and 6, so $q^2(1+q^2)^2 = 1$, which reduces to

$$q^3 + q = 1, \qquad q = 0.68233, \qquad \rho_{\mathrm{biphenylene}} = 0.18883. \tag{9}$$

The setting that reaches the threshold is the one in the proof: the six bonds of the minority structure grow by 18.9% and the other eight shrink by 18.9% (Fig. 1b).

A second argument certifies ρ for any skeleton. Each Kekulé structure uses each bond at most once, so $\det T$ is an affine function of every single coupling. Its smallest value over the box of allowed couplings is therefore reached at a corner of the box, where each coupling sits at $1+\epsilon$ or $1-\epsilon$. Checking the sign of $\det T$ at all 2^m corners, with m the number of bonds, just below and just above ρ proves the threshold.

3 Exact computation of the threshold

The closure threshold of Eq. (5) is the regularity radius of the interval matrix $[T - \epsilon T,\, T + \epsilon T]$: the smallest ϵ at which the set of matrices contains a singular one [15–17]. For general matrices it is NP-hard to compute [15]. Corner enumeration costs 2^m determinants, which is 4×10^6 for a [3]phenylene and 10^9 for a [4]phenylene.

Poljak and Rohn showed that only 2^{2n} special corners matter. For sign vectors $y, z \in \{-1, 1\}^n$, change coupling t_{ij} by $-\epsilon\, y_i z_j\, t_{ij}$. The determinant along this family vanishes at $\epsilon = 1/\lambda$ for each real eigenvalue λ of $T^{-1}(T \circ yz^{\mathsf{T}})$, where $\circ$ is the elementwise product, and

$$\rho = \Big[\max_{y,z}\ \lambda_{\mathbb{R}}\big(T^{-1}(T \circ yz^{\mathsf{T}})\big) \Big]^{-1}, \tag{10}$$

Table 1: Closure thresholds. K: number of Kekulé structures; ASC: algebraic structure count; g: Hückel gap at equal couplings; ρ: closure threshold from Eq. (10); last column: smallest single-bond change that closes the gap (+, increase; −, decrease), or none while all couplings stay non-negative.

skeleton	atoms	K	ASC	$g/\lvert\beta\rvert$	ρ	single bond
benzene	6	2	2	2.000	–	none
naphthalene	10	3	3	1.236	–	none
benzocyclobutadiene	8	3	1	0.525	0.1197	−50%
biphenylene	12	5	3	0.890	0.1888	+300%
linear [3]phenylene	18	12	4	0.525	0.1197	−100%
angular [3]phenylene	18	13	5	0.844	0.1962	+500%
linear [4]phenylene	24	29	5	0.348	0.0756	−83%
bent [4]phenylene	24	31	7	0.565	0.1265	+233%
helical [4]phenylene	24	34	8	0.703	0.1614	+400%
zigzag [4]phenylene	24	34	8	0.743	0.1614	+400%
triangular [4]phenylene	24	35	9	0.905	0.2031	+900%

with $\lambda_{\mathbb{R}}$ the largest absolute value of a real eigenvalue [15, 17]. Fixing $z_1 = 1$ removes the overall sign and leaves 2^{2n-1} eigenvalue problems of size n. The cost grows with the number of atoms and not with the number of bonds. For the four smallest skeletons with a minority sign, Eq. (10) agrees with corner enumeration to 10^{-10}.

Table 1 lists the thresholds for the [N]phenylenes with $N \leq 4$. Every phenylene closes with a cooperative change between 7.6% and 20.3%, while the single-bond route needs a coupling to grow severalfold or to shrink to half or less. Along the linear series the threshold falls from 18.9% to 12.0% to 7.6%, the same direction as the rising reactivity and red shift of linear phenylenes [2, 3]. Angular and branched skeletons need larger changes than linear ones of the same size. Helical and zigzag [4]phenylene share the same threshold to all digits although their gaps differ (0.703 and 0.743). They are a pair of the kind described in Section 2: a graph-isomorphism test on the incidence between bonds and signed Kekulé structures finds a sign-preserving correspondence of their bonds, so their determinants are the same polynomial in the couplings.

Exactness matters in practice. A search that starts from random sign vectors and flips one sign at a time until Eq. (10) stops improving returns 0.1666 for zigzag [4]phenylene, above the exact value 0.1614.

4 Closing patterns and closing orbitals

The threshold is reached at a *closing pattern*: every coupling at $1 + \rho$ or $1 - \rho$. When one Kekulé structure carries the minority sign, the proof of the Proposition identifies the pattern: its bonds strengthen and all others weaken. For the larger phenylenes, which have several minority structures, Eq. (10) returns the pattern through the maximizing y and z.

At a closing pattern T has left and right null vectors ϕ and ψ, one on each sublattice. They define the *closing orbital*, the pair of orbitals that reach zero energy; atom i carries weight $\phi_i^2/2$ on one sublattice and $\psi_j^2/2$ on the other. Figure 2 shows these weights for six phenylenes.

The skeleton decides where closure happens. In linear phenylenes the closing orbital spreads along the whole chain, with its weight peaking at the centre. In angular and branched skeletons it concentrates at one end. In angular [3]phenylene one terminal benzene–cyclobutadiene–benzene

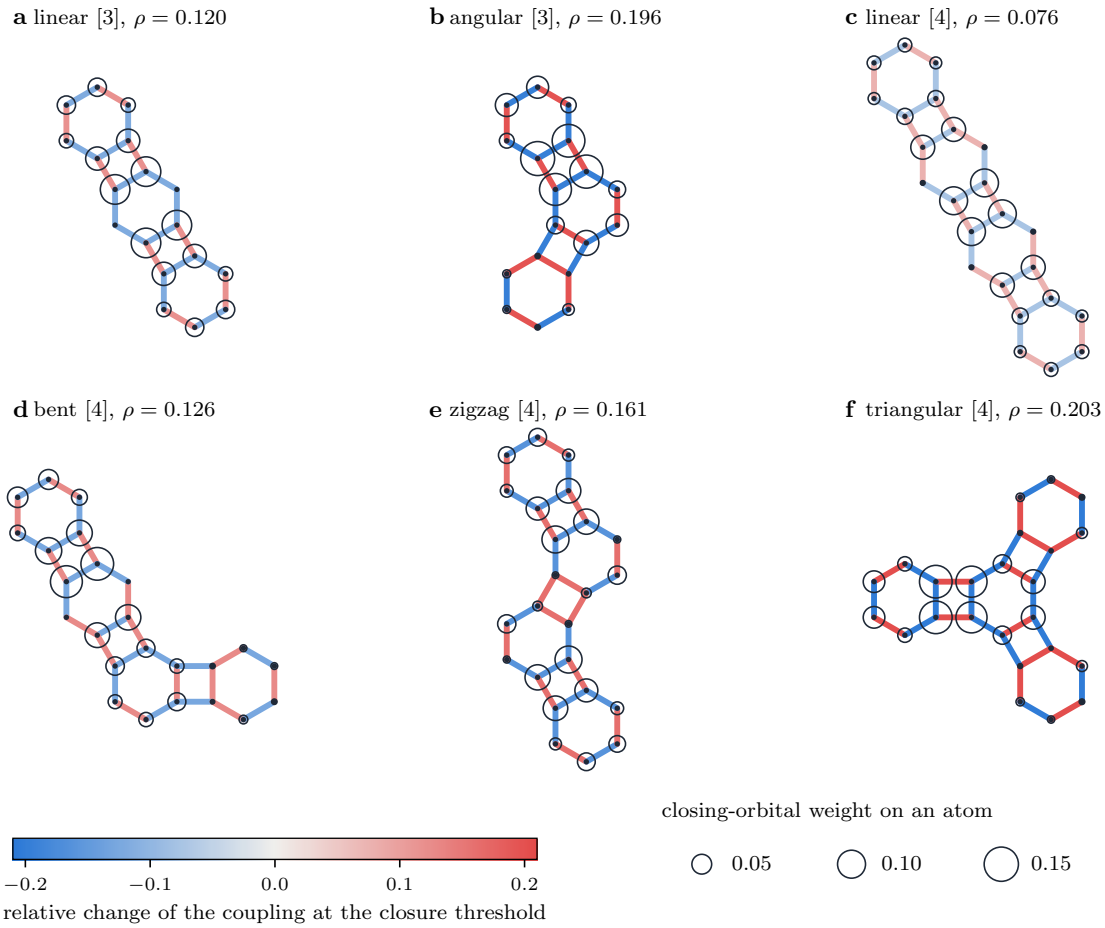


Figure 2: Closing patterns and closing orbitals of six [N]phenylenes, drawn to a common scale. Bond colour: relative change of the coupling at the closure threshold (red, stronger; blue, weaker). Circle area: weight of the closing orbital on each atom.

unit carries 90% of the weight; in bent [4]phenylene one terminal unit carries 68%; in triangular [4]phenylene the central ring and one arm carry 82%. In zigzag and helical [4]phenylene the two terminal units carry nearly all of it and the central four-membered ring 2%.

Is the end that closes a choice of the algorithm? Collecting every sign-vector pair that attains the maximum in Eq. (10) lists all closing patterns. Angular [3]phenylene has exactly two, up to overall sign, and they are mirror images: one closes the gap at one end, the other at the opposite end. Bent [4]phenylene has one; triangular [4]phenylene has three, related by its threefold rotation, each on one arm; zigzag and helical [4]phenylene have one each, shared by both ends. No closing pattern of an angular or branched skeleton spreads over the molecule. Closure there is local, with as many equivalent sites as the symmetry allows.

Linear [3]phenylene shows the opposite case, a whole-chain closure inherited from a smaller molecule. Its threshold and its gap both equal those of benzocyclobutadiene (Table 1). The mirror plane perpendicular to the long axis passes through the two central-ring atoms that do not belong to a four-membered ring. Orbitals that are antisymmetric under this mirror vanish on these two atoms, which disconnects the two halves; each half is benzocyclobutadiene. The spectrum of benzocyclobutadiene is therefore part of the spectrum of linear [3]phenylene, including its frontier

pair. Applying the closing pattern of benzocyclobutadiene to both halves closes the antisymmetric block, so $\rho_{\text{lin-3}} \le \rho_{\text{BCB}}$; the exact computation gives equality.

5 Closure energy in a bond-length model

A molecule changes its couplings by changing bond lengths, and bond lengths resist. We let each coupling depend on the change δ_b of its bond length from a common reference, $t_b = t_0 - \alpha\,\delta_b$, and add a harmonic σ energy [18]:

$$E(\delta) = -2\sum_k \sigma_k\big(T(\delta)\big) + \sum_b \Big(\tfrac{k_\sigma}{2}\delta_b^2 + c\,\delta_b\Big), \qquad c = -2\alpha p_{\text{ref}}. \tag{11}$$

With $p_{\text{ref}} = 2/3$, benzene is undistorted at $\delta = 0$, and at equilibrium $\delta_b = 2\alpha(p_{\text{ref}} - p_b)/k_\sigma$, a linear relation between bond length and bond order p_b. We use $t_0 = 2.5$ eV and $\alpha = 4.1$ eV Å^{-1} [18] and a σ force constant $k_\sigma = 43$ eV Å^{-2}, which gives a slope of 0.19 Å per unit bond order; $k_\sigma = 21$ and 60 eV Å^{-2} give the same ranking of all results below (Appendix A).

Relaxing the bond lengths widens the gap in all nine skeletons: the molecule moves away from closure. We define the *closure energy* ΔE as the lowest energy we find on the set of geometries with $\det T(\delta) = 0$, measured from the relaxed geometry, using constrained optimization from many starting points. It ranges from 1.9 eV for linear [4]phenylene to 5.5 eV for triangular [4]phenylene (Table 2). The topological threshold already ranks these energies: ρ, which knows nothing about bond lengths, has Spearman correlation 0.94 with ΔE and places biphenylene (0.189) before angular [3]phenylene (0.196), as ΔE does.

What sets ΔE? Near the relaxed geometry the gap changes linearly and the energy quadratically,

$$g(x) \approx g_0 + a^{\mathsf{T}}x, \qquad E(x) - E_0 \approx \tfrac{1}{2}x^{\mathsf{T}}Fx, \tag{12}$$

with a the gradient of the gap with respect to bond lengths and F the Hessian of the energy. Minimizing the energy subject to $g = 0$ gives

$$\Delta E \approx \frac{g_0^2}{2\,a^{\mathsf{T}}F^{-1}a}, \qquad x^* = -\frac{g_0\,F^{-1}a}{a^{\mathsf{T}}F^{-1}a}. \tag{13}$$

The denominator measures how much gap one buys per unit of deformation energy. We call x^* the *predicted pattern*.

The numerator comes from the frontier orbitals. Let u and v be the singular vectors of the relaxed T for $\sigma_{\min}$. Then $\partial g/\partial t_{ij} = 2u_iv_j$. In orbital language the HOMO is $(u, v)/\sqrt{2}$ and the LUMO $(u, -v)/\sqrt{2}$, so $u_iv_j = p_{ij}^{\text{H}} - p_{ij}^{\text{L}}$, the difference between the HOMO and LUMO partial bond orders, the classical first-order response of an orbital energy to a coupling [4]. We write

$$P = \sum_{(ij)}\big(p_{ij}^{\text{H}} - p_{ij}^{\text{L}}\big)^2 = \sum_{(ij)}(u_iv_j)^2, \tag{14}$$

so that $a^{\mathsf{T}}a = 4\alpha^2 P$. P is large when the frontier orbitals concentrate on few bonds.

The denominator is almost a constant. $F = k_\sigma I + \alpha^2\,\partial^2 E_\pi/\partial t^2$, and the second term is the bond–bond polarizability [4, 5]. Across the nine skeletons the eigenvalues of F lie between 27.9 and 43.0 eV Å^{-2}. Setting $F \approx \kappa I$ in Eq. (13) gives

$$\Delta E \approx \frac{\kappa}{8\alpha^2}\,\frac{g_0^2}{P}. \tag{15}$$

Table 2: Closure energy in the bond-length model ($k_\sigma = 43$ eV Å^{-2}). ρ: closure threshold (Table 1); g_0: gap of the relaxed geometry; P: Eq. (14); ΔE_{loc}: Eq. (13); ΔE: lowest closure energy found. Last rows: Spearman rank correlation with ΔE and its two-sided p value from the exact permutation distribution (all 9! orderings; ties as mid-ranks).

skeleton	ρ	g_0 (eV)	P	g_0^2/P	ΔE_{loc} (eV)	ΔE (eV)
linear [4]phenylene	0.076	1.35	0.268	6.79	1.93	1.87
benzocyclobutadiene	0.120	2.09	0.644	6.78	2.00	2.01
linear [3]phenylene	0.120	1.75	0.326	9.37	2.67	2.70
bent [4]phenylene	0.126	1.77	0.282	11.16	3.11	3.05
biphenylene	0.189	2.58	0.449	14.87	4.31	4.18
helical [4]phenylene	0.161	2.08	0.212	20.40	5.49	4.51
zigzag [4]phenylene	0.161	2.20	0.229	21.22	5.93	4.52
angular [3]phenylene	0.196	2.39	0.274	20.83	5.63	4.77
triangular [4]phenylene	0.203	2.48	0.226	27.34	7.48	5.55
Spearman with ΔE	0.94	0.72		0.97	0.98	
p	5.5×10^{-4}	0.037		1.7×10^{-4}	5.0×10^{-5}	

Equation (15) explains why the gap of the relaxed geometry ranks closure poorly (Table 2). Biphenylene has a larger gap than angular [3]phenylene, 2.58 against 2.39 eV, but its frontier orbitals concentrate on fewer bonds ($P = 0.45$ against 0.27), and its gap closes for 4.18 instead of 4.77 eV. Over the nine skeletons the rank correlations with ΔE are 0.98 for Eq. (13), 0.97 for g_0^2/P, 0.94 for ρ, 0.90 for the gap at equal couplings and 0.72 for the gap of the relaxed geometry. Relaxation widens the gaps by different amounts and scrambles their order; ρ and g_0^2/P follow the closure energy. All of these numbers come from one bond-length model, so they measure how well its topological and local approximations describe it; whether real molecules behave the same way is the subject of Section 6. Equation (13) reproduces ΔE within 5% for benzocyclobutadiene, biphenylene, the linear skeletons and bent [4]phenylene and overestimates it by 18–35% for the angular and branched ones, whose closures lie farthest from the relaxed geometry.

The two factors come from different orbital pairs. The gap gradient depends only on the HOMO and LUMO. The bond–bond polarizability is a sum over occupied–virtual pairs of products of transition terms $c_i^{\mathrm{occ}} c_j^{\mathrm{vir}} + c_j^{\mathrm{occ}} c_i^{\mathrm{vir}}$, and for the HOMO–LUMO pair of an alternant hydrocarbon this term vanishes on every bond by the pairing theorem [7, 19]. The pair that controls the gap does not enter the stiffness.

6 A preregistered test in real molecules

Does the predicted pattern of a crude model still lower the gap efficiently in a real molecule, and does the reversal between biphenylene and angular [3]phenylene in Table 2 carry over to their open-shell character? Before computing any displaced geometry we fixed the directions, the controls, the observables and two pass criteria in a file whose SHA-256 hash is reported with the data (Appendix B).

Protocol. Both molecules were optimized with B3LYP/def2-SVP [20–22] in PySCF [23]. The equilibrium gaps are 4.10 eV (biphenylene) and 3.75 eV (angular [3]phenylene). Three kinds of bond-length change were applied, all scaled to the same model energy $\frac{1}{2} x^{\mathsf{T}} F x$: the predicted pattern x^*; a *bond-equalizing control* that undoes the relaxed bond-length alternation of the model; and

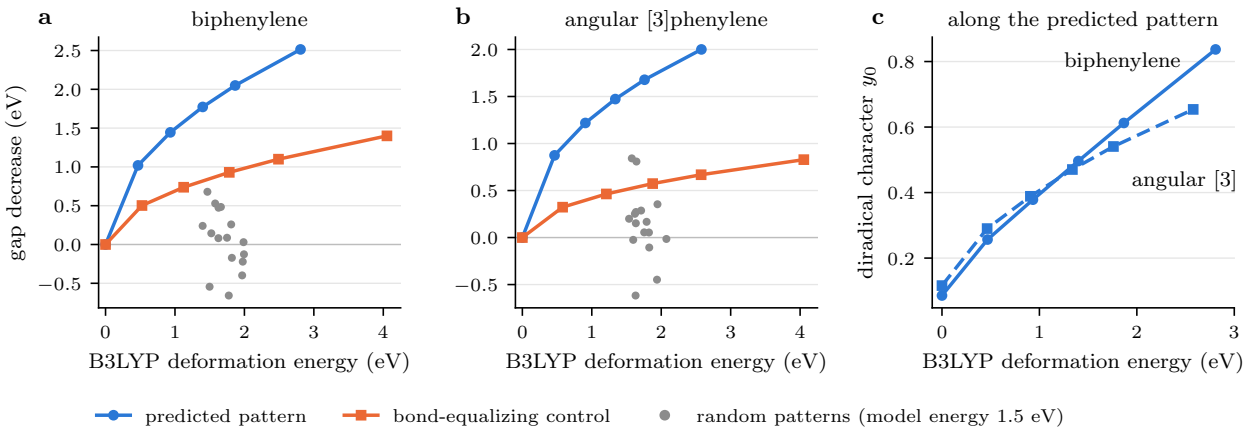


Figure 3: B3LYP test. (a, b) Decrease of the gap against B3LYP deformation energy for the predicted pattern, the bond-equalizing control and the random patterns (model energy 1.5 eV). (c) Diradical character along the predicted pattern of each molecule.

eight random patterns, each with both signs. Bond-length changes were converted to in-plane Cartesian displacements of the carbon atoms with the Wilson B matrix [24], hydrogens following their carbons. With fewer C–C bonds than in-plane carbon coordinates the solution is not unique; we took the minimum-norm least-squares solution and iterated it to convergence. Every pattern is realized exactly, the bond-length changes matching the targets to 10^{-15} Å; the bond angles adjust, and their strain enters the B3LYP energy. At each geometry, without further relaxation, we computed the closed-shell (restricted) B3LYP energy and the Kohn–Sham HOMO–LUMO gap of this restricted solution; throughout, "gap" means this orbital gap and deformation energies are differences of restricted energies. The broken-symmetry singlet lies below the restricted one only beyond a model energy of 1.5 eV: by at most 0.02 eV at 2 eV and by 0.08–0.19 eV at 3 eV. Criterion 1 is therefore unaffected, and the deformation energies quoted below for open-shell onsets and crossings, all at or below 2 eV, would move by at most 0.02 eV with broken-symmetry energies. Along the predicted pattern we also computed the diradical character $y_0 = 1 - 2\tau/(1+\tau^2)$, where τ is half the difference between the occupations of the highest occupied and lowest unoccupied UHF natural orbitals [25, 26], and the spin contamination $\langle S^2 \rangle$ of the broken-symmetry UB3LYP singlet.

Criterion 1: the predicted pattern lowers the gap most efficiently. At a model energy of 1.5 eV we compared the efficiency η, the gap decrease per eV of B3LYP deformation energy, with the median over the eight random patterns, each taken with its better sign. The criterion required a ratio of at least 1.5 in both molecules. The ratios are 5.5 for biphenylene ($\eta = 1.27$ against 0.23) and 6.6 for angular [3]phenylene ($\eta = 1.10$ against 0.17), and no random pattern reaches half the efficiency of the predicted one (Fig. 3). The threshold of 1.5 was low: when the response is isotropic, the best possible direction beats random ones in 14 and 22 dimensions by a median ratio of 5.3 and 6.7 when eight random directions are drawn (computed by sampling; the large-d limit $\sqrt{d}/0.67$ gives 5.6 and 7.0). Two comparisons carry more information. In the model itself, the predicted pattern beats the same eight random patterns by factors of 6.7 and 8.7; B3LYP retains 82% and 76% of this advantage. And the predicted pattern lowers the gap 2.4 and 3.6 times more efficiently than the bond-equalizing control ($\eta = 0.52$ and 0.31), the physically natural alternative.

Criterion 2: biphenylene becomes open-shell first. The criterion required a larger y_0 for biphenylene at every matched deformation energy and an earlier onset of spin symmetry breaking ($\langle S^2 \rangle > 0.1$). Neither holds. Angular [3]phenylene starts with the larger diradical character, 0.116

against 0.086, keeps it up to 1.17 eV, and breaks spin symmetry first, at 1.76 against 1.87 eV. The criterion was stricter than the model. With the numbers of Table 2, Eq. (13) predicts the gap of angular [3]phenylene to stay below that of biphenylene up to 0.66 eV of deformation and to be overtaken only then. B3LYP shows this crossing at larger deformation: y_0 of biphenylene overtakes at 1.17 eV, and its gap, 0.34 eV higher at the start, falls below that of angular [3]phenylene at 2.0 eV. We report the crossing as an exploratory result outside the preregistered criteria. In both molecules spin symmetry breaks while the B3LYP gap is still 2.0–2.1 eV.

The predicted pattern therefore carries over to the real molecules as a direction, and the faster closing of biphenylene shows up in both the gap and y_0, later than the model predicts. Following the fixed criteria, we read the test as supporting the mechanism of Section 5; the crossing in open-shell order is consistent with the model but was not part of what we set out to test.

7 Discussion

The ingredients of the energy of closing are classical. The gap gradient is the difference of frontier partial bond orders [4], a first-order property of the kind for which rigorous bounds have recently been derived [27]; the stiffness combines σ springs with the bond–bond polarizability that governs bond fixation [5] and symmetry lowering by π electrons [6, 7]; and the competition between electronic driving force and stiffness is the pseudo-Jahn–Teller picture [28]. What this paper adds is the finite question: how large a change of the couplings closes the gap, where it must sit, and whether the predicted pattern survives in a real molecule. The threshold follows exactly from the cancellation of Kekulé structures and has a closed form when one structure has the minority sign. Closure concentrates at one end of angular phenylenes and is shared by the chain in linear ones. Under B3LYP the predicted pattern keeps about 80% of its model advantage over random patterns and outperforms the bond-equalizing control 2.4- to 3.6-fold. Because ρ depends only on the signed Kekulé structures, it ranks closure before any bond length is known, and g_0^2/P explains the ranking once bond lengths are included.

Two questions follow naturally. The first is how closely the Hückel gap gradient matches the B3LYP one bond by bond; computing the B3LYP gradient in bond coordinates would turn the 80% retention into an angle between the two predicted patterns. The second is whether the efficiency advantage decides which molecule becomes open-shell first when two molecules start with the same diradical character; pairs of isomers with matched y_0 but different P would answer it.

The Proposition covers every skeleton with one minority Kekulé structure, and it would be useful to know which skeletons these are. For larger skeletons, Eq. (10) grows as 4^n; grouping sign vectors related by the symmetry of the skeleton gains a constant factor, and the planarity of molecular graphs may allow faster exact methods.

A Bond-length model

Parameters: $t_0 = 2.5$ eV, $\alpha = 4.1$ eV Å^{-1}, $k_\sigma \in \{21, 43, 60\}$ eV Å^{-2}, $p_{\mathrm{ref}} = 2/3$. Relaxed geometries were found by BFGS from $\delta = 0$. Closure energies were obtained by sequential quadratic programming with the constraint $\det T(\delta)/t_0^n = 0$, whose gradient follows from the adjugate. Starting points were the closing pattern of Eq. (10), the relaxed geometry plus multiples of it, and random perturbations of the relaxed geometry; the lowest feasible result is reported. Closure energies at $k_\sigma = 21$, 43 and 60 eV Å^{-2} range over 1.7–2.9, 1.9–5.5 and 2.2–7.7 eV and rank the nine skeletons identically. For triangular [4]phenylene the relaxed frontier level is doubly degenerate; there a in Eq. (13) is taken in the direction of the two-dimensional frontier subspace that maximizes $a^{\mathsf{T}} F^{-1} a$, and P likewise.

B DFT protocol and deviations

All calculations used B3LYP/def2-SVP with density fitting in PySCF 2.14; geometries were optimized with pyberny. Displaced geometries were single points. The broken-symmetry singlet was converged from a guess that mixes HOMO and LUMO and from the UHF solution, keeping the lower energy. UHF was converged from a HOMO–LUMO mixing guess and from a guess with opposite spin density on the two carbon sublattices, keeping the lower energy. The preregistration file and its change log (with SHA-256 hashes after each change) are provided with the data. Four changes were made during the work, each recorded before the affected results were used:

1. y_0 was taken from UHF natural orbitals instead of CASSCF(2,2), which was too slow for angular [3]phenylene; at the biphenylene reference CASSCF(2,2) gives $y_0 = 0.058$.
2. In a first pass, UHF for angular [3]phenylene converged to a nearly restricted solution from the HOMO–LUMO guess alone. The sublattice guess was added and both molecules were recomputed; it lowers the UHF energy of the angular [3]phenylene reference by 1.13 eV and leaves the biphenylene results unchanged. This change was made after the first part of the biphenylene ladder and the angular [3]phenylene reference had been seen.
3. Stability analysis of the UHF and UKS solutions was dropped as too slow and replaced by the two starting guesses above.
4. Open-shell quantities were computed only along the predicted pattern; they do not enter criterion 1.

The criteria and their thresholds were not changed.

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Author contributions Xiyao Yu conceived the study, developed the mathematical framework, performed the analytical and computational investigations, analyzed the results, and wrote and revised the manuscript.

Data and code availability The source code, computational results, validation protocol and change log, and data underlying all figures are publicly available at https://github.com/PillowSoprano/phenylene-gap-closure. The code is released under the MIT License, and the data and figures under CC BY 4.0.

Ethics approval Not applicable. This study involved theoretical analysis and computational simulations and did not involve human participants or animals.

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